
Handwriting Author(?) Identification



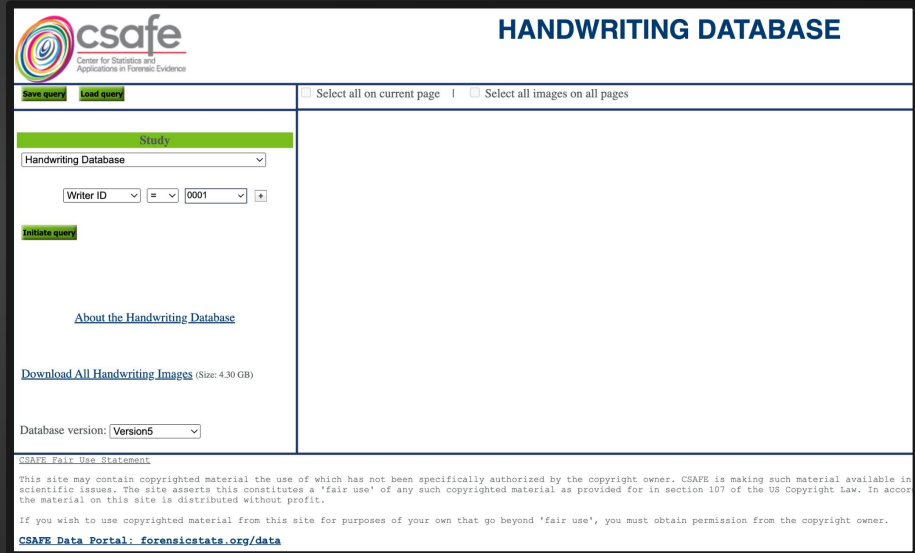
Zainab Siddiqui
Data Capstone I



Recap

→ CSAFE's Handwriting Database

- ◆ CSAFE: Center for Statistics and Applications in Forensic Evidence
 - Handwriting Database, version 5
- ◆ Image data
 - Grayscale
 - ~2521x2810
- ◆ 475 classes x 27 images per class = **12,825 instances**
- ◆ 15/6/6 train/val/test split
- ◆ **CSV file with writer info*****



The screenshot shows the CSAFE Handwriting Database interface. At the top, the CSAFE logo is on the left, and the title "HANDWRITING DATABASE" is on the right. Below the logo, there are "Save query" and "Load query" buttons. To the right of these buttons are two checkboxes: "Select all on current page" and "Select all images on all pages". The main content area is divided into two columns. The left column contains a "Study" dropdown menu set to "Handwriting Database", a "Writer ID" dropdown set to "0001", and an "Initiate query" button. Below this is a link "About the Handwriting Database" and another link "Download All Handwriting Images (Size: 4.30 GB)". At the bottom of the left column is a "Database version:" dropdown set to "Version5". The right column is empty. At the very bottom of the page, there is a "CSAFE Fair Use Statement" and a link to the "CSAFE Data Portal: forensicatata.org/data".

Our London business is good, but
Vienna and Berlin are quiet..

Mr. D. Lloyd has gone to Switzerland
And I hope for good news. He will
be there for a week at 1496 Zornott
Street and then goes to Turin
and Rome and will join Colonel
Parry and arrive at Athens Greece
November 27 or December 2. Letters

there should be addressed

King James Blvd 3580, We expect

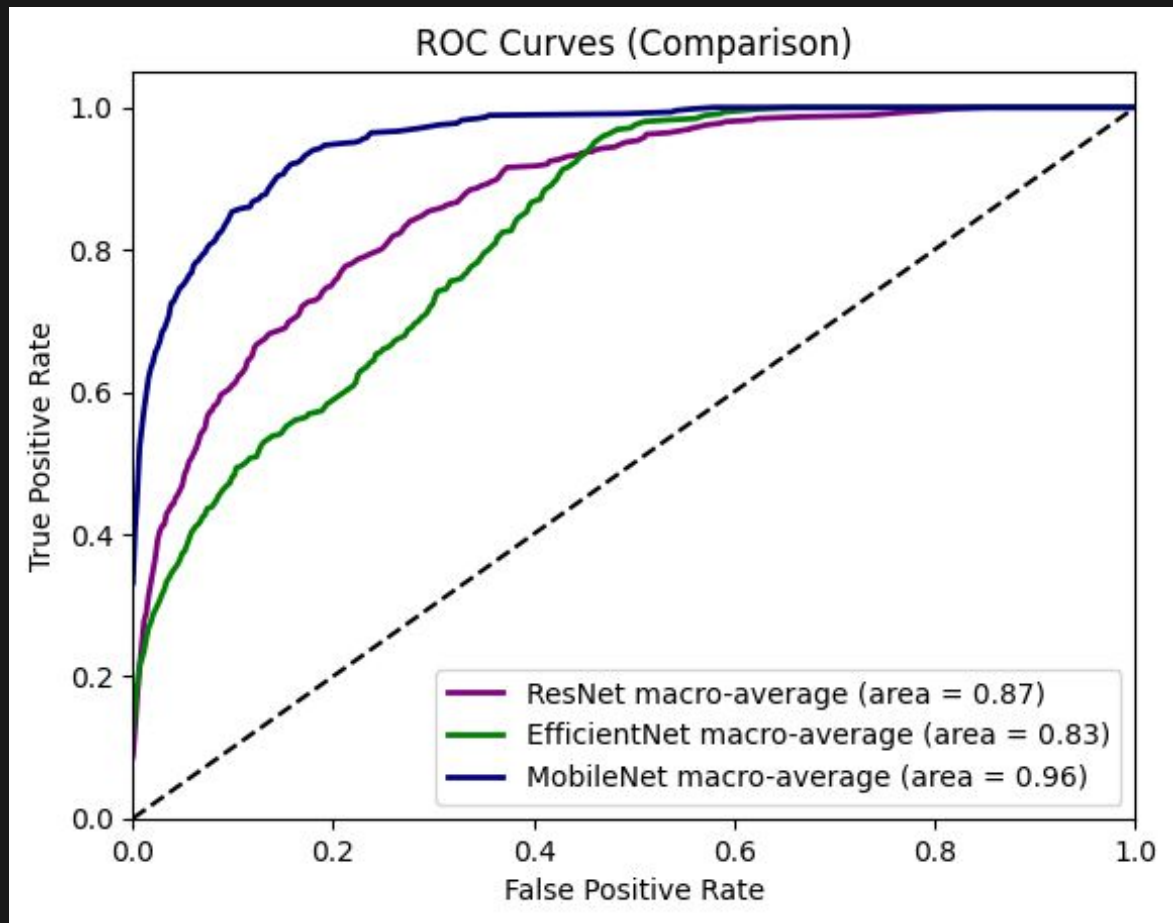
Charles F. Fuller Tuesday

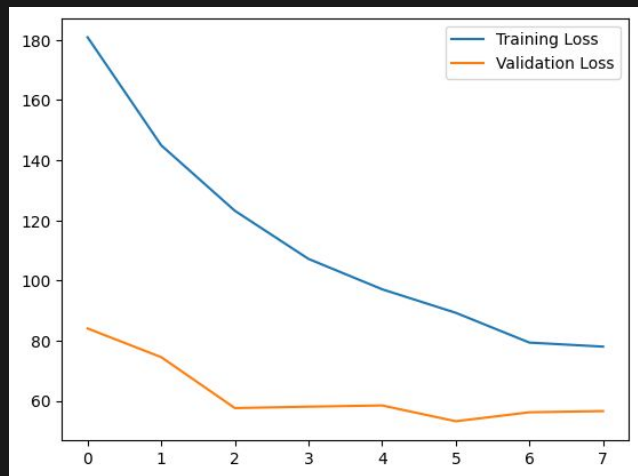
Dr. McQuaid and Robert Unger
Esq. left on the 'Y.X.' Express
tonight

Within a short time she was walking briskly
toward the Emerald City, her silver shoes tinkling
merely on the hard, yellow roadbed. The
sun shone bright and the birds sang sweet
and Dorothy did not feel nearly as bad as
you might think a little girl would who had
been suddenly whisked away from her own
country and set down in the midst of a
strange land.

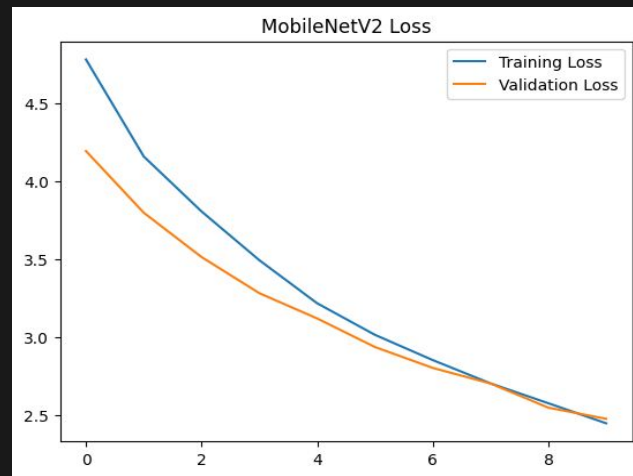
The early bird may get the worm, but
the second mouse gets the cheese.

Baseline

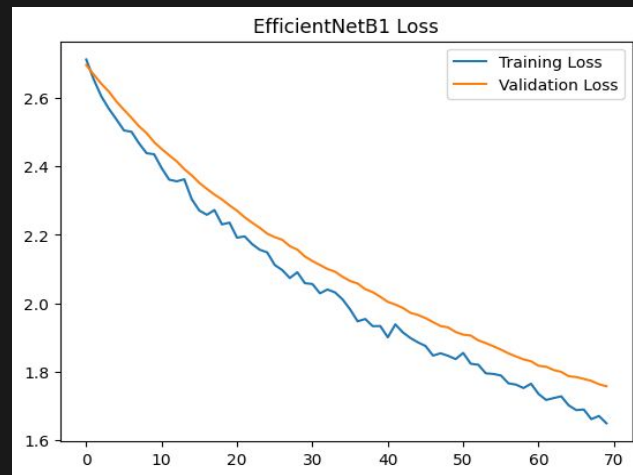
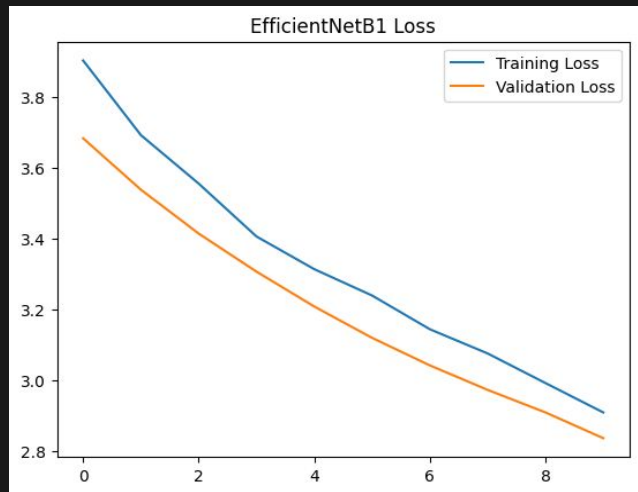




✦ ResNet50V2



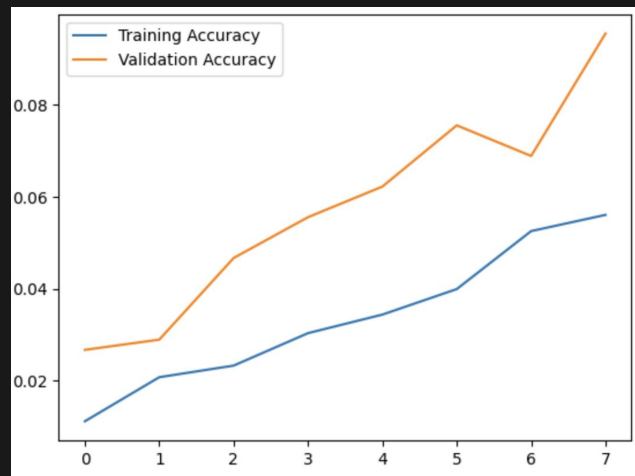
Loss Graphs



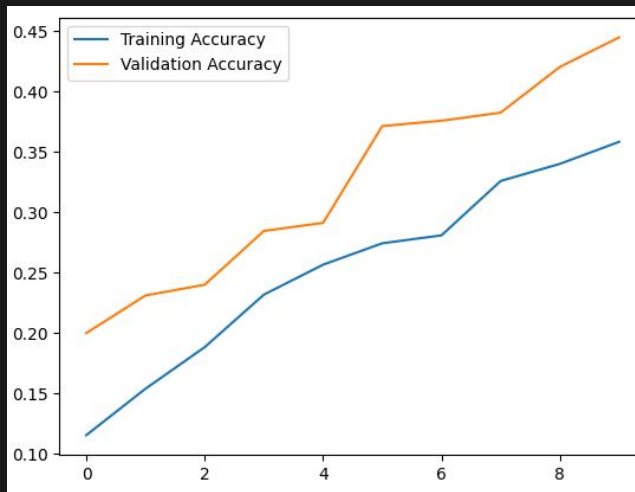
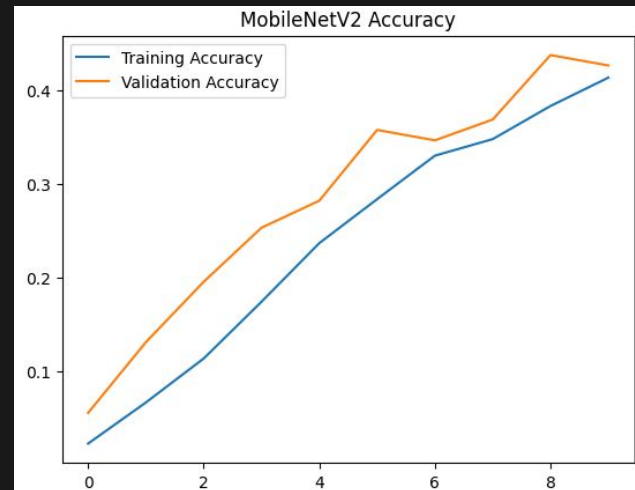
Most recent ✦

70 epochs ✦

70% raw accuracy ✦

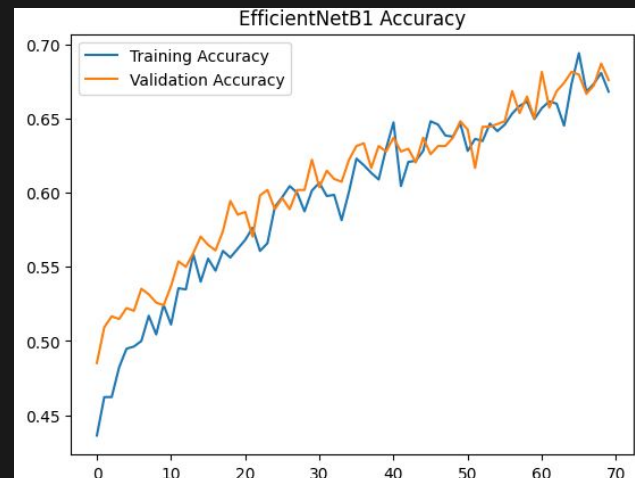


✦ ResNet50V2



✦ EfficientNetB1
(original)

Accuracy Graphs



Test Data Example

- ◆ Each number is a different class/writer
- ◆ ~70% accuracy

```
IMAGE_SIZE = (442,442)
BATCH_SIZE = 64
INPUT_SHAPE = IMAGE_SIZE + (3,)
NUM_CLASSES = 90
```

T: w0131
P: w0074

The camp bird may get the worm, but the second mouse gets the cheese.

T: w0025
P: w0025

Our London business is good, but Vienna and Berlin are quiet. Mr. D. Lloyd has gone to Switzerland and I hope the good news. He will be there for a week at 1486 Zermatt Street and then goes to Turin and Rome and will join Colonel Perry and arrive at Athens, Greece, November 27 or December 2. Letters there should be addressed King James Blvd, 3850. We expect Charles E. Fuller Tuesday, Dr. L. McQuaid and Robert Unger, Eng, left on the 'Y.K.' Express tonight.

T: w0155
P: w0022

Our London business is good, but Vienna and Berlin are quiet. Mr. D. Lloyd has gone to Switzerland and I hope the good news. He will be there for a week at 1486 Zermatt Street and then goes to Turin and Rome and will join Colonel Perry and arrive at Athens, Greece, November 27 or December 2. Letters there should be addressed to King James Blvd, 3850. We expect Charles E. Fuller Tuesday, Dr. L. McQuaid and Robert Unger, Eng, left on the 'Y.K.' Express tonight.

T: w0022
P: w0022

Let's have a short time, she was talking, suddenly turned the round city for silver shoes, suddenly moving on the hard, yellow mud. The sun shows bright and the birds sang sweet and dwelling did not feel nearly as bad as you might think, a little girl would who had been suddenly whisked away from her own country and let down in the midst of a strange land.

T: w0029
P: w0029

Our London business is good, but Vienna and Berlin are quiet. Mr. D. Lloyd has gone to Switzerland and I hope the good news. He will be there for a week at 1486 Zermatt Street and then goes to Turin and Rome and will join Colonel Perry and arrive at Athens, Greece, November 27 or December 2. Letters there should be addressed King James Blvd, 3850. We expect Charles E. Fuller Tuesday, Dr. L. McQuaid and Robert Unger, Eng, left on the 'Y.K.' Express tonight.

T: w0076
P: w0076

Our London business is good, but Vienna and Berlin are quiet. Mr. D. Lloyd has gone to Switzerland and I hope the good news. He will be there for a week at 1486 Zermatt Street and then goes to Turin and Rome and will join Colonel Perry and arrive at Athens, Greece, November 27 or December 2. Letters there should be addressed King James Blvd, 3850. We expect Charles E. Fuller Tuesday, Dr. L. McQuaid and Robert Unger, Eng, left on the 'Y.K.' Express tonight.

T: w0027
P: w0027

The camp bird may get the worm, but the second mouse gets the cheese.

T: w0078
P: w0078

Thrifty bird, may get the worm, but the second mouse gets the cheese.

T: w0073
P: w0043

While in each time the sun setting, brightly through the forest. Day, up when day, bright, mostly in the back, yellow, under the sun. Sun shows bright and the birds sang sweet and dwelling did not feel nearly as bad as you might think, a little girl would who had been suddenly whisked away from her own country and let down in the midst of a strange land.

→ Next Challenge



◆ Majority class: 18-24 (~50%)

◆ Majority class: Female (~65%)

Size: 466 x 13

	wid	agegroup	gender	hand	thirdgrade_usa	thirdgrade_usa_region	s1_dae	s1_time	s2_dae	s2_time	s3_dae	s3_time
0	180	61+	Female	Right	True	Midwest	0.0	Late Morning	28.0	Late Morning	66.0	Late Morning
1	410	18-24	Female	Right	True	Midwest	0.0	Late Afternoon	32.0	Late Evening	54.0	Early Evening
2	593	18-24	Male	Left	True	Midwest	0.0	Late evening (later than 7:30pm)	30.0	Late evening (later than 7:30pm)	61.0	Early evening (5pm-7:30pm)
3	520	18-24	Female	Right	True	Midwest	0.0	Late Evening	26.0	Late Morning	54.0	Late Morning
4	233	18-24	Male	Right	False	NaN	340.0	Late Evening	380.0	Late Evening	0.0	Late Evening
5	35	41-60	female	right	False	NaN	0.0	lateevening	24.0	latemorning	64.0	earlyevening
6	661	18-24	Female	Right	True	Midwest	0.0	Late Afternoon	41.0	Late Afternoon	57.0	Late Evening
7	420	18-24	Female	Right	True	NaN	0.0	Late Afternoon	52.0	Early Afternoon	73.0	Late Afternoon
8	62	41-60	female	right	True	midwest	0.0	lateevening	54.0	latemorning	98.0	earlyevening
9	572	18-24	Female	Right	True	Midwest	0.0	Late morning (9:30am-12:00pm)	24.0	Late morning (9:30am-12:00pm)	54.0	Early afternoon (12:00pm-2:30pm)

◆ Majority class: Right handedness (~87%)



Questions?
